Progressive Feedforward Collapse of ResNet Training

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***Abstract*—Neural collapse (NC) is a simple and symmetric phenomenon for deep neural networks (DNNs) at the terminal phase of training, where the last-layer features collapse to their class means and form a simplex equiangular tight frame (ETF) aligning with the classifier vectors. However, the relationship of the last-layer features to the data and intermediate layers during training remains unexplored. To this end, we characterize the geometry of intermediate layers of residual neural network (ResNet) and propose a novel conjecture, *progressive feedforward collapse* (PFC), claiming the degree of collapse increases during the forward propagation of DNNs. We derive a transparent model for the well-trained ResNet based on the principle that ResNet with weight decay approximates the geodesic curve in the Wasserstein space at the terminal phase. The metrics of PFC indeed monotonically decrease across depth on various datasets. We propose a new surrogate model, *multilayer unconstrained feature model* (MUFM), connecting intermediate layers by an optimal transport regularizer. The optimal solution of MUFM is inconsistent with NC but is more concentrated relative to the input data. Overall, this study extends NC to PFC to model the collapse phenomenon of intermediate layers and its dependence on the input data, shedding light on the theoretical understanding of ResNet in classification problems.**

***Index Terms*—Deep learning, geodesic curve, neural collapse (NC), optimal transport, residual neural network (ResNet), Wasserstein space.**

I. Introduction

DEEP neural networks (DNNs) have been applied to diverse fields and achieved remarkable success in computer vision [1], natural language processing [2], bioinformatics [3], and so on. One way to explain the power of DNNs is to discover some specific phenomena in deep learning training and theoretically analyze them. For example, overparameterized DNNs can achieve zero training error even for random label data. Zhang et al. [4] illustrated this phenomenon by a theoretical construction of a two-layer network. More generally, overparameterized DNNs exhibit a "double descent" phenomenon that the test error decreases twice as model complexity increases [5]. Lee and Cherkassky [6] analyzed this under the VC-theoretical framework. The information bottleneck (IB) theory reveals that the stochastic gradient descent (SGD) optimization has two main phases, i.e., empirical error minimization and representation compression, and the converged layers are closed to the IB bound [7], [8], which has motivated to apply the IB principle to the DNN training [9], [10], [11]. Implicit regularization shows that, although there are lots of global minima due to overparameterization, DNNs trained by SGD still converge to the max-margin solutions without any explicit regularizer [12], [13], [14].

Recently, Papyan et al. [15] revealed the neural collapse (NC) phenomenon for a DNN classification task on a balanced dataset, where the last-layer features and classifier vectors exhibit a simple and highly symmetric geometry at the terminal phase of training. Specifically, the features collapse to their corresponding class means, while the class means (centered at the global mean) converge to a simplex equiangular tight frame (ETF). Concurrently, the classifier vectors align with the centered class means. As a consequence of this geometry, the classifier converges to choose whichever class has the nearest training class mean. Notably, this phenomenon has been observed across various DNNs employed for the classification tasks [16].

However, NC solely characterizes the last-layer features and the classifier. To gain a more comprehensive understanding of DNNs, we aim to investigate the geometry of intermediate layers. How do DNNs transform the input data to the final simplex ETF during the forward propagation? Several studies have indicated that the features become progressively more concentrated and separated across layers, accompanied by a decrease in feature variability through layers [17], [18], [19], [20], [21], [22], [23], [24], [25]. Furthermore, examining this problem from a dynamical viewpoint during forward propagation contributes to a clear understanding. Haber and Ruthotto [26] and E [27] first interpreted DNNs with ordinary differential equations. In particular, the residual neural network (ResNet) [1] stands out as a widely adopted architecture with rich theoretical insights [28], [29], [30], [31], [32]. Gai and Zhang [28] established a connection between the

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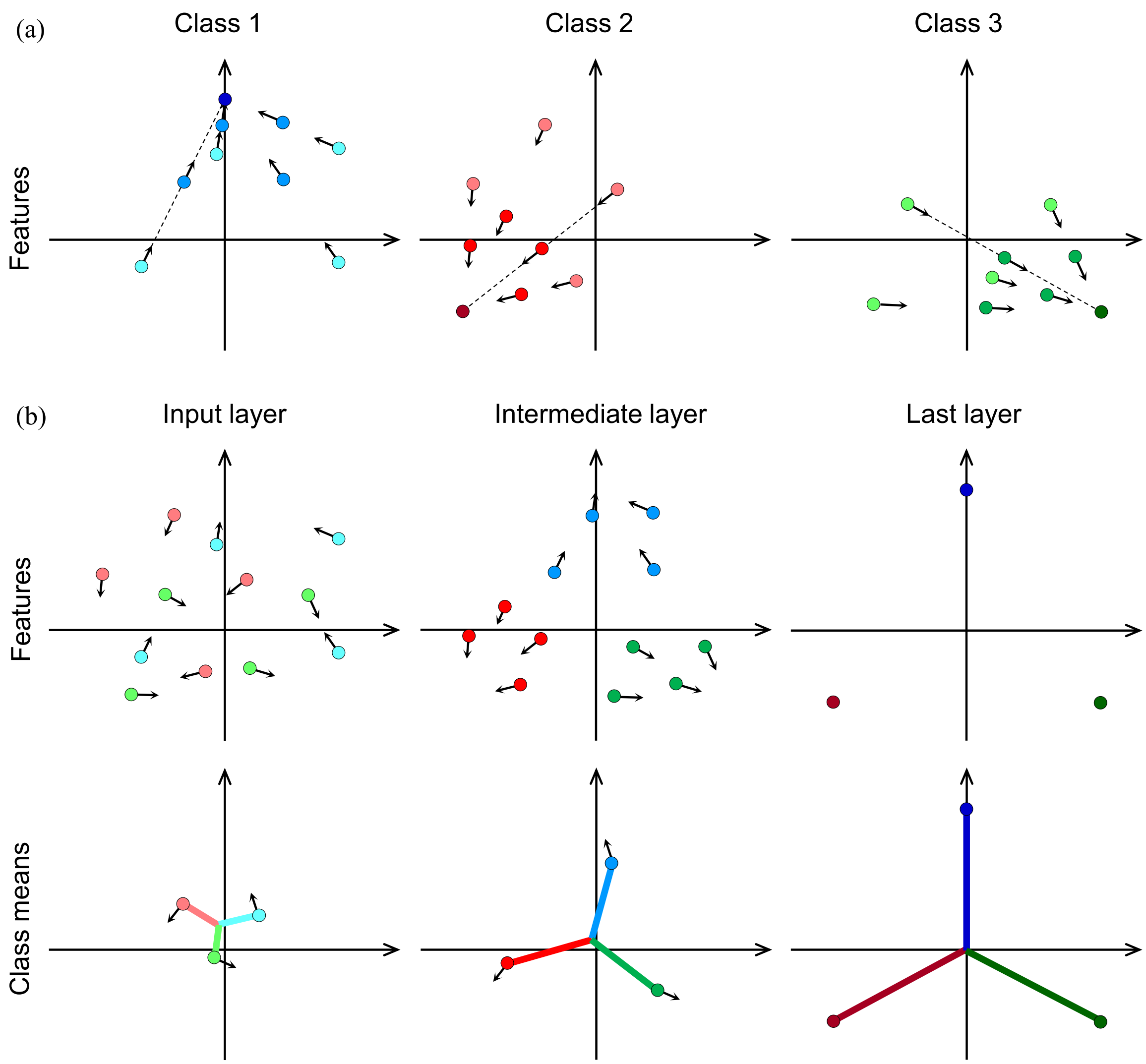

Fig. 1. Illustration of PFC. This 2-D example represents a three-class dataset depicted by different colors. Each class consists of four data points. Under the geodesic curve assumption, these points undergo uniform linear motion during forward propagation in ResNet, whose color gradually darkens. The arrow indicates the direction of the motion. (a) Trajectory of features for each class during forward propagation in ResNet. The dotted line shows the motion trajectory. (b) Distribution of the input (left). Distribution of the intermediate layer (middle). Distribution of the last layer, exhibiting NC (right). Trajectory of features during forward propagation in ResNet (top). The features progressively collapse to their respective class means. Trajectory of (centered) class means (bottom). The class means, centered at the global mean, progressively collapse to the simplex ETF (see Definition 1).

forward propagation of ResNet and the continuity equation, demonstrating that ResNet with weight decay attempts to learn the geodesic curve in the Wasserstein space. In other words, the forward propagation of ResNet is approximately along a straight line at the terminal phase of training, and we call this the geodesic curve assumption.

In this article, we investigate the intermediate layers of ResNet from the perspective of NC and introduce a novel conjecture named progressive feedforward collapse (PFC) (Fig. 1). In short, for a well-trained ResNet, the degree of collapse increases during the forward propagation. We empirically show that the PFC conjecture is valid on various datasets, as all metrics associated with PFC decrease through layers at the final epoch. To understand the PFC phenomenon, we incorporate the geodesic curve assumption to model the forward propagation and predict the features of intermediate layers. This suggests that ResNets transform the input data to the simplex ETF along a straight line, and the features become more collapsed in deeper layers. Theoretically, we can prove that the PFC metrics decrease monotonically, while the features collapse to the final simplex ETF along this line under some mild assumptions. To make our theoretical analysis more closely aligned with the neural networks used in practice, we prove that when the NC and the geodesic curve assumption are not perfectly satisfied, the PFC1 metric mentioned below still decreases monotonically most of the time in the forward dynamic modeling of ResNet. We also extend the PFC conjecture on vision transformer (ViT) [33] to investigate more network architectures with residual connection. Additionally, the attention layer contributes more significantly to classification (indicated by the PFC metrics) compared to the MLP layer in ViTs.

To better illustrate the PFC phenomenon, we propose a novel surrogate model termed the multilayer unconstrained feature model (MUFM) to capture both NC and PFC. Previous studies have theoretically analyzed NC using a simplified model known as the unconstrained feature model (UFM) or layer-peeled model [16], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43]. This model treats the last-layer features as free variables and optimizes the classifier and features simultaneously, achieving global optimality being consistent with NC. However, UFM neglects the relationship between the input data and intermediate features, presenting an overidealized scenario. The input data can determine the last-layer features in NC [44]. In addition, achieving perfect NC (e.g., all metrics of NC reach zero) is impractical. To connect the input data and last-layer features, the MUFM model treats the intermediate-layer features as optimization variables through the optimal transport regularizer, serving as a lower bound of weight decay [28]. The last-layer features are not treated as free variables but are related to the data. We demonstrate that the optimal solution of MUFM deviates from the NC solution but exhibits a higher concentration than the initial data. Moreover, there is a tradeoff between the data and the simplex ETF by controlling the coefficient of the optimal transport regularizer.

This article is organized as follows. We provide the notations, background of NC, and theoretical understanding of NC and ResNet in Section II. In Section III, we propose the PFC conjecture and elucidate it under the geodesic curve assumption. Subsequently, we propose MUFM to understand both NC and PFC. In Section IV, we provide some empirical evidence for the conjecture and results mentioned in Section III. In Section V, we list some related works. Finally, we provide conclusions and discussions in Section VI.

II. Preliminaries

In this article, we focus on DNNs trained on balanced datasets. We consider a K-class classification problem, where each class comprises n training samples, i.e., nK samples in total. Denote $\boldsymbol{x}_{k,i} \in \mathbb{R}^D$ as the ith training sample of the kth class. Its corresponding label is the one-hot vector $\boldsymbol{y}_k \in \mathbb{R}^K$. Formally, DNN can be written as

$$f_{\boldsymbol{\Theta}}(\boldsymbol{x}) = \boldsymbol{W}\boldsymbol{h}_{\boldsymbol{\theta}}(\boldsymbol{x}) + \boldsymbol{b}$$
$$\boldsymbol{h}_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(\boldsymbol{W}_L(\cdots\sigma(\boldsymbol{W}_1\boldsymbol{x}+\boldsymbol{b}_1)) + \boldsymbol{b}_L)$$

where $\boldsymbol{h}_{\boldsymbol{\theta}}(\cdot) : \mathbb{R}^D \to \mathbb{R}^d$ is the feature mapping and $\sigma(\cdot)$ is an element-wise nonlinear function. $\boldsymbol{W} = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_K]^\top \in \mathbb{R}^{K\times d}$ and $\boldsymbol{b} \in \mathbb{R}^K$ are the classifier matrix and bias. Specifically, we assume that $d > K$, which is a common scenario in practice. $\boldsymbol{\Theta} = \{\boldsymbol{W}, \boldsymbol{b}, \boldsymbol{\theta}\}$ is the set of all DNN parameters.

Our goal is to train the parameters $\boldsymbol{\Theta}$ by minimizing the empirical risk over all training samples

$$\min_{\boldsymbol{\Theta}} \frac{1}{Kn}\sum_{k=1}^{K}\sum_{i=1}^{n}\mathcal{L}\left(f_{\boldsymbol{\Theta}}\left(\boldsymbol{x}_{k,i}\right), \boldsymbol{y}_k\right) + \mathcal{R}(\boldsymbol{\Theta})$$

where $\mathcal{L}(\cdot,\cdot)$ is the loss function, e.g., cross-entropy (CE) or mean squared error (MSE), and $\mathcal{R}(\cdot)$ is the regularizer (e.g., L_2-norm regularization).

For $l = 1, \ldots, L$, let us denote the l-layer feature of the ith training sample of the kth class by $\boldsymbol{h}^l_{k,i}$ and the last-layer features by $\{\boldsymbol{h}_{k,i}\}$ [i.e., $\boldsymbol{h}_{k,i} = \boldsymbol{h}^L_{k,i} = \boldsymbol{h}_{\boldsymbol{\theta}}(\boldsymbol{x}_{k,i})$]. The class mean and global mean of the l-layer features are $\boldsymbol{h}^l_k$ and $\boldsymbol{h}^l_G$

$$\boldsymbol{h}^l_k = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{h}^l_{k,i}$$
$$\boldsymbol{h}^l_G = \frac{1}{K}\sum_{k=1}^{K}\boldsymbol{h}^l_k = \frac{1}{nK}\sum_{k=1}^{K}\sum_{i=1}^{n}\boldsymbol{h}^l_{k,i}.$$

The l-layer features within-class covariance $\boldsymbol{\Sigma}^l_W$ and between-class covariance $\boldsymbol{\Sigma}^l_B$ are computed by

$$\boldsymbol{\Sigma}^l_W = \frac{1}{nK}\sum_{k=1}^{K}\sum_{i=1}^{n}\left(\boldsymbol{h}^l_{k,i} - \boldsymbol{h}^l_k\right)\left(\boldsymbol{h}^l_{k,i} - \boldsymbol{h}^l_k\right)^\top$$
$$\boldsymbol{\Sigma}^l_B = \frac{1}{K}\sum_{k=1}^{K}\left(\boldsymbol{h}^l_k - \boldsymbol{h}^l_G\right)\left(\boldsymbol{h}^l_k - \boldsymbol{h}^l_G\right)^\top.$$

A. Neural Collapse

Papyan et al. [15] revealed the NC phenomenon for a DNN classification task on balanced datasets, where the last-layer features and classifier vectors exhibit a simple and highly symmetric geometry at the terminal phase of training. We introduce the simplex ETF first.

Definition 1 (Simplex ETF) A simplex ETF is a collection of vectors $\boldsymbol{m}_i \in \mathbb{R}^d, i = 1, 2, \ldots, K, d \geq K-1$, specified by the columns of

$$\boldsymbol{M} = \sqrt{\frac{K}{K-1}}\boldsymbol{U}\left(\boldsymbol{I}_K - \frac{1}{K}\boldsymbol{1}_K\boldsymbol{1}_K^T\right) \tag{1}$$

where $\boldsymbol{U} \in \mathbb{R}^{d\times K}$ is a partial orthogonal matrix and satisfies $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}_K, \boldsymbol{I}_K \in \mathbb{R}^{K\times K}$ is the identity matrix, and $\boldsymbol{1}_K \in \mathbb{R}^K$ is the all ones vector.

Formally, NC can be described with four properties.

(NC1) Variability Collapse: The within-class variability converges to zero, which means that the last-layer features of each class collapse to their class means during the training, i.e.,

$$\boldsymbol{\Sigma}_W \to \boldsymbol{0}.$$

(NC2) Convergence to the Simplex ETF: The class means centered at the global mean converge to the simplex ETF (see Definition 1). That is to say, the centered class means have equal length and equal, maximally separated pair-wise angles, i.e.,

$$\frac{\langle \boldsymbol{h}_k - \boldsymbol{h}_G, \boldsymbol{h}_{k'} - \boldsymbol{h}_G\rangle}{\|\boldsymbol{h}_k - \boldsymbol{h}_G\|_2\|\boldsymbol{h}_{k'} - \boldsymbol{h}_G\|_2} \to \begin{cases} 1, & k = k' \\ \dfrac{-1}{K-1}, & k \neq k' \end{cases}$$
$$\|\boldsymbol{h}_k - \boldsymbol{h}_G\|_2 - \|\boldsymbol{h}_{k'} - \boldsymbol{h}_G\|_2 \to 0 \quad \forall k \neq k'.$$

(NC3) Convergence to the Self-Duality: The classifier vectors are in alignment with the centered class means, i.e.,

$$\frac{\boldsymbol{w}_k}{\|\boldsymbol{w}_k\|_2} - \frac{\boldsymbol{h}_k - \boldsymbol{h}_G}{\|\boldsymbol{h}_k - \boldsymbol{h}_G\|_2} \to 0 \quad \forall k.$$

(NC4) Simplification to the Nearest Class Center (NCC): The classifier converges to choosing whichever class has the

nearest training class mean. Given a new feature $\boldsymbol{h}$, it is classified by

$$\arg\max_k \langle \boldsymbol{w}_k, \boldsymbol{h} \rangle + b_k \to \arg\min_k \|\boldsymbol{h} - \boldsymbol{h}_k\|_2 .$$

The NC phenomenon only characterizes the geometry of the last-layer features. It is natural to investigate the behavior of intermediate layers. Several works have shown that the features become more and more concentrated and separated through layers [17], [18], [19]. Furthermore, many empirical findings indicate that the variability collapse (NC1) emerges at the intermediate layers but is weaker than the last layer [20], [21], [22], [23], [24], [25]. Ben-Shaul and Dekel [22] and Galanti et al. [23] also provided a similar result on NC4. To characterize NC at the intermediate layers more carefully, we introduce the two terms, i.e., the NCC *separability* and *empirical effective depth* proposed in [23].

The l-layer features are NCC separable means that the true class of each sample can be identified correctly by finding the nearest class mean of the corresponding feature. Formally, for $\forall k \in [K], i \in [n]$, we have

$$\arg\min_{c\in[K]} \left\| \boldsymbol{h}_{k,i}^l - \boldsymbol{h}_c^l \right\|_2 = k.$$

Next, we use the empirical effective depth to denote the lowest depth that satisfies the NCC separability in practice. The empirical result shows that if the L_0-layer features are NCC separable, all l-layer ($l \geq L_0$) features are NCC separable. We allow a small error ϵ when measuring the NCC separability empirically. Consider a L-hidden layer well-trained neural network, and the ϵ-empirical effective depth is the minimal $l \in [L]$ that satisfies

$$\frac{1}{nK}\sum_{k=1}^{K}\sum_{i=1}^{n} \mathbb{I}\left[\arg\min_{c\in[K]} \left\| \boldsymbol{h}_{k,i}^l - \boldsymbol{h}_c^l \right\|_2 \neq k\right] \leq \epsilon.$$

Recently, several works characterized the intermediate NC properties after the effective depth, e.g., the layers after a certain layer exhibit the NC properties [45], [46]. We mainly focus on the layers before the effective depth. As an intriguing phenomenon, several works also investigated some implications of NC for DNNs' performance [47], [48], [49], [50], [51] (see a review for more details [52]).

B. Understanding NC by a UFM

It is challenging to characterize the NC phenomenon due to the high nonconvexity of DNNs with interaction between different layers. Thus, previous works only explored a simplified model (i.e., a UFM or a layer-peeled model) instead of DNNs.

UFM treats the last-layer features as free optimization variables and neglects the intermediate layers, so the DNN is reduced to a linear model whose variables are the last-layer features and the classifier vectors. This simplification is based on the fact that DNNs are highly overparameterized and can approximate a large family of functions. We can focus on the last-layer features $\boldsymbol{H}$ and the classifier $\boldsymbol{W}$ to study their optimality. Researchers mainly study the global optimality, loss landscape, and training dynamics in different settings of the UFM, such as different loss functions (e.g., MSE and CE), regularizers (weight decay), or constraints on $\boldsymbol{W}$ and $\boldsymbol{H}$. Here, we introduce the UFM under the CE loss and weight decay on $\boldsymbol{W}$ and $\boldsymbol{H}$.

Let $\boldsymbol{W} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_K]^\top \in \mathbb{R}^{K\times d}$ and $\boldsymbol{H} = [\boldsymbol{h}_{1,1}, \dots, \boldsymbol{h}_{1,n}, \boldsymbol{h}_{2,1}, \dots, \boldsymbol{h}_{K,n}] \in \mathbb{R}^{d\times Kn}$ be the matrices of the classifier and last-layer features. UFM is defined as follows:

$$\min_{\boldsymbol{W},\boldsymbol{H}} \frac{1}{Kn}\sum_{k=1}^{K}\sum_{i=1}^{n} \mathrm{CE}\left(\boldsymbol{W}\boldsymbol{h}_{k,i}, \boldsymbol{y}_k\right) + \frac{\lambda_{\boldsymbol{W}}}{2}\|\boldsymbol{W}\|_F^2 + \frac{\lambda_{\boldsymbol{H}}}{2}\|\boldsymbol{H}\|_F^2 \quad (2)$$

where CE indicates the CE loss and $\lambda_{\boldsymbol{W}}$ and $\lambda_{\boldsymbol{H}} > 0$ are the hyperparameters for weight decay.

Although the problem is nonconvex, recent works proved that the global minimizers are indeed NC solutions [35], [36], [37], [38], [39], [40], [41], [42], [43]. Moreover, some studies [40], [41], [42], [43] showed that UFM has a benign loss landscape, that is, there is no spurious local minimizer on the landscape, which means that all local minima are global minima. Therefore, any optimization methods that can escape strict saddle points (e.g., SGD) will converge to the NC solutions. Han et al. [16] and Mixon et al. [34] studied the gradient flow of UFM and proved that it converges to the NC solutions under some assumptions. Beyond UFM, some studies extended it to two or arbitrary layers and proved the global optimality [39], [46], [53], which is closer to the practical DNNs.

Overall, as a simplified model of DNNs, UFM gives a theoretical understanding of NC but is still limited. This simplification breaks the relation between the data and the learned features, and the regularizer or constraint on $\boldsymbol{H}$ is not used in practice. The regularizer on $\boldsymbol{H}$ is interpreted as upper bounded by the weight decay on all layers if the inputs are bounded.

C. ResNet Learns the Geodesic Curve in the Wasserstein Space

To explore more layers and take forward propagation into account, we introduce the dynamical system view of deep learning in this section. Gai and Zhang [28] built the connection of ResNet and the continuity equation, showing that ResNet with weight decay attempts to learn the geodesic curve in the Wasserstein space. Thus, the forward propagation of ResNet is along a straight line.

The connection between ResNet and the dynamical system was first introduced in [26] and [27]. The general form of a ResNet block [54] in forward propagation is

$$\boldsymbol{h}^{(l+1)} = \boldsymbol{h}^{(l)} + \boldsymbol{v}_l\left(\boldsymbol{h}^{(l)}\right), \quad l \in \{0, 1, \dots, L-1\} \quad (3)$$

where $\boldsymbol{v}_l(\cdot)$ is a function induced by a shallow network and $\boldsymbol{h}^{(l)}$ is the input of the lth ResNet block. It can be viewed as a discretization of the dynamical system

$$\frac{d\boldsymbol{h}}{dt} = \boldsymbol{v}(\boldsymbol{h}, t), \quad t \in [0, 1]. \quad (4)$$

Denote the data distribution, target distribution, and the lth ResNet block output distribution by μ_0, μ_1, and μ_l, respectively. Then, the corresponding dynamical system has the form

$$\frac{d\mu_t}{dt} = \tilde{v}(\mu_t, t), \quad t \in [0, 1]. \quad (5)$$

Gai and Zhang [28] claimed that the dynamical system (5) should conserve mass, corresponding to the continuity equation, which has the form

$$\frac{d\mu_t}{dt} + \nabla \cdot (v_t \mu_t) = 0, t \in [0, 1] \tag{6}$$

where the vector field v_t is the infinitesimal variation of the continuum μ_t.

When μ_0 and μ_1 are fixed, there are infinite curves in the probability measure space connecting μ_0 and μ_1 by continuity equations. It has been shown that a special one, the geodesic curve in the Wasserstein space, has good generalization ability. It is induced by the optimal transport map T from μ_0 to μ_1 with the minimum cost. This cost is called the Wasserstein distance $W_c(\mu_0, \mu_1)$, i.e.,

$$W_c(\mu_0, \mu_1) = \inf_{T_{\#}\mu_0 = \mu_1} \int c(\boldsymbol{x}, T(\boldsymbol{x}))\, d\mu_0(x)$$

where $c(\cdot, \cdot)$ is the cost function of transporting mass from μ_0 to μ_1 and $T_{\#}$ is the push-forward operator. The cost function is usually set to $\|\boldsymbol{x} - T(\boldsymbol{x})\|_2^2$ in both optimal transport and deep learning. In this setting, the geodesic curve (μ_t) is induced by the optimal transport map T, i.e.,

$$\mu_t = ((1 - t) I_d + tT)_{\#} \mu_0.$$

For each point $\boldsymbol{x} \in \operatorname{supp}(\mu_0)$, $\boldsymbol{x}$ is transported along a straight line $\boldsymbol{x}^{(t)} = (1 - t)\boldsymbol{x} + tT(\boldsymbol{x})$. Thus, we want to get the optimal transport map T by the neural network, and then, the forward propagation is along a straight line. The following proposition provides a way to compute the Wasserstein distance.

Proposition 1 (Benamou–Brenier Formula *[55]*) Let μ_0, $\mu_1 \in \mathcal{P}(\mathbb{R}^d)$. Then, it holds

$$W_2^2(\mu_0, \mu_1) = \inf \left\{ \int_0^1 \|v_t\|_{L^2(\mu_t)}^2 \, dt \right\}$$

where the infimum is taken among all weakly continuous distributional solutions of the continuity equation (μ_t, v_t) connecting μ_0 and μ_1.

Gai and Zhang [28] proved that the weight decay is an upper bound of the above energy in DNNs. In addition, the forward propagation of ResNet satisfies the continuity equation. According to the Benamou–Brenier formula, ResNet with weight decay aims to approximate the geodesic curve in the Wasserstein space. Thus, we propose the geodesic curve assumption.

Assumption 1 (Geodesic Curve Assumption) At the terminal phase of training, the ResNet with weight decay has learned the geodesic curve in the Wasserstein space $\mathcal{P}(\mathbb{R}^d)$, which is induced by the optimal transport map. As a result, the forward propagation of ResNet is along a straight line in the feature space $\mathbb{R}^d$.

III. Progressive Feedforward Collapse

NC reveals a distinctive geometric structure in the last-layer features of DNNs. It is interesting to investigate the behavior of features in the intermediate layers, particularly how neural networks like ResNet transform the initial data to the simplex ETF during forward propagation. This process is crucial for gaining insights into classification tasks. While previous studies have shown that the features tend to become increasingly concentrated and separated through layers [17], [18], [19], [20], [21], [22], [23], [24], [25], a clear and transparent model of the entire network, analogous to the NC phenomenon observed in the last layer, is still lacking. To address this issue, in this section, we mainly consider the well-trained ResNet and propose a novel conjecture named PFC.

A. PFC Conjecture

To investigate the behavior of features in the intermediate layers, we first extend the statistics used for NC to the intermediate layers. For the features of each layer, we compute the following metrics to measure the degree of collapse.

The variability collapse of the layer l

$$\mathcal{PFC}_1\left(\boldsymbol{H}^l\right) := \frac{\mathbb{E}\left[\left\|\boldsymbol{h}_{k,i}^l - \boldsymbol{h}_k^l\right\|_2^2\right]}{\mathbb{E}\left[\left\|\boldsymbol{h}_k^l - \boldsymbol{h}_G^l\right\|_2^2\right]} = \frac{\operatorname{Tr}\left(\boldsymbol{\Sigma}_W^l\right)}{\operatorname{Tr}\left(\boldsymbol{\Sigma}_B^l\right)}. \tag{7}$$

We use the same metric in [20] and [24], which is a little different from the original NC1 metric $\operatorname{Tr}(\boldsymbol{\Sigma}_W \boldsymbol{\Sigma}_B^{\dagger})$ [15], where $\boldsymbol{\Sigma}_B^{\dagger}$ denotes the pseudoinverse of $\boldsymbol{\Sigma}_B$. The numerator of $\mathcal{PFC}_1^{(l)}$ is the within-class variance and the denominator is the between-class variance. $\mathcal{PFC}_1^{(l)}$ is more amenable for analysis. It can still capture the behavior of variability collapse that is $\boldsymbol{\Sigma}_W \to \boldsymbol{0}$, while $\boldsymbol{\Sigma}_B \not\to \boldsymbol{0}$ is nondegenerate.

The distance between the l-layer (centered) class means and the simplex ETF

$$\mathcal{PFC}_2\left(\boldsymbol{H}^l\right) := \left\| \frac{\tilde{\boldsymbol{H}}^{l\top} \tilde{\boldsymbol{H}}^l}{\left\|\tilde{\boldsymbol{H}}^{l\top} \tilde{\boldsymbol{H}}^l\right\|_F} - \boldsymbol{E} \right\|_F \tag{8}$$

where $\tilde{\boldsymbol{H}}^l := [\boldsymbol{h}_1^l - \boldsymbol{h}_G^l, \ldots, \boldsymbol{h}_K^l - \boldsymbol{h}_G^l] \in \mathbb{R}^{d \times K}$ is the centered class mean matrix and $\boldsymbol{E} = 1/((K-1)^{1/2})(\boldsymbol{I}_K - 1/K \boldsymbol{1}_K \boldsymbol{1}_K^\top)$. We extend the metric in [40] to each layer, considering the geometry of simplex ETF with a single metric.

The NCC accuracy of the layer l

$$\mathcal{PFC}_3\left(\boldsymbol{H}^l\right) := \frac{1}{nK} \sum_{k=1}^{K} \sum_{i=1}^{n} \mathbb{I}\left[\operatorname*{arg\,min}_{c \in [K]} \left\|\boldsymbol{h}_{k,i}^l - \boldsymbol{h}_c^l\right\|_2 = k\right]. \tag{9}$$

$\mathcal{PFC}_3^{(l)}$ measures the NCC separability of each layer. When $\mathcal{PFC}_3^{(l)} = 1$, we say that the features of the layer l are NCC separable. Intuitively, the features are more NCC separable when they are closer to their class mean and the centered class means are closer to the simplex ETF.

At the terminal phase of training, the last-layer features collapse to their class means and the class means (centered at global mean) converge to the simplex ETF. Meanwhile, DNNs transform the initial data to the simplex ETF during forward propagation. The metrics mentioned above can be employed to quantify the degree of collapse. By computing these metrics at each layer, we can investigate the geometry of intermediate layers and observe the variation trend. As a result, we can conjecture that the following holds.

Conjecture 1 (PFC Conjecture) *At the terminal phase of ResNet training, NC emerges at the last-layer features. There exists an order in the degree of collapse (measured by three PFC metrics) of each layer before the effective depth. Concretely, the following conditions hold.*

(PFC1) The features of each layer progressively collapse to their class means.
(PFC2) The centered class means of each layer feature progressively collapse to the simplex ETF.
(PFC3) The NCC accuracy of each layer progressively collapses to 1.

During the training process, the metrics at each layer start at a random initialization and gradually converge to the final order.

The PFC conjecture characterizes the geometrical variation in the forward propagation of ResNet. It shows that each ResNet block makes the features more concentrated to their class means and the class means more separated. Thus, ResNet progressively extracts features during forward propagation that make the data classified more easily. We give partial proof of this conjecture under the geodesic curve assumption next and some empirical evidence in Section IV. To investigate more network architectures with residual connection, we also provide some empirical evidence for PFC in ViTs and their variants in Section IV and Appendix (Supplementary Material).

B. Monotonicity Under the Geodesic Curve Assumption

In Section II-C, we introduce the geodesic curve assumption, that is, the forward propagation of ResNet is along a straight line. Formally, for each data point $\boldsymbol{x}_{k,i}$, the feature map $\boldsymbol{h}_{k,i}^{(t)}$ in the forward propagation of ResNet with $t \in [0, 1]$ is along the line $\boldsymbol{h}_{k,i}^{(t)} = (1-t)\boldsymbol{h}_{k,i}^{(0)} + t\boldsymbol{h}_{k,i}^{(1)}$. This assumption models the forward propagation and can help to reveal how ResNet transforms the initial data to the simplex ETF. PFC shows that the features progressively collapse to the final simplex ETF. In this section, we prove that under the geodesic curve assumption, PFC metrics monotonically decrease across depth at the terminal phase of training, which explains the PFC phenomenon. The detailed proofs are provided in Appendixes A-C (Supplementary Material).

Assumption 2: Suppose the last-layer features $\{\boldsymbol{h}_{k,i}^{(1)}\}$ exhibit NC, that is, $\boldsymbol{h}_{k,i}^{(1)} = \boldsymbol{h}_{k}^{(1)}\ \forall i, k$, and the set of vectors $\{\boldsymbol{h}_{k}^{(1)} - \boldsymbol{h}_{G}^{(1)}\}$ composes the simplex ETF.

Then, under the geodesic curve assumption, the forward propagation of ResNet is

$$\begin{aligned} \boldsymbol{h}_{k,i}^{(t)} &= (1-t)\,\boldsymbol{h}_{k,i}^{(0)} + t\boldsymbol{h}_{k}^{(1)}, \quad t \in [0,1] \\ \boldsymbol{h}_{k}^{(t)} &= (1-t)\,\boldsymbol{h}_{k}^{(0)} + t\boldsymbol{h}_{k}^{(1)}, \quad t \in [0,1] \\ \boldsymbol{h}_{G}^{(t)} &= (1-t)\,\boldsymbol{h}_{G}^{(0)} + t\boldsymbol{h}_{G}^{(1)}, \quad t \in [0,1] \end{aligned} \tag{10}$$

and the forward propagation of the centered class mean matrix of ResNet $\tilde{\boldsymbol{H}}(t) := [\boldsymbol{h}_1^{(t)} - \boldsymbol{h}_G^{(t)}, \ldots, \boldsymbol{h}_K^{(t)} - \boldsymbol{h}_G^{(t)}]$ is

$$\tilde{\boldsymbol{H}}(t) = (1-t)\,\tilde{\boldsymbol{H}}(0) + t\tilde{\boldsymbol{H}}(1), \quad t \in [0,1]. \tag{11}$$

We also need an assumption over the start and end points.

Assumption 3: Assume that $\sum_{k=1}^{K}\langle \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_G^{(0)}, \boldsymbol{h}_k^{(1)} - \boldsymbol{h}_G^{(1)}\rangle \geq 0$.

Remark 1: Given the input $\{\boldsymbol{h}_k^{(0)}\}$, ResNet transforms $\{\boldsymbol{h}_{k,i}^{(0)}\}$ to $\{\boldsymbol{h}_k^{(1)}\}$ as an optimal transport map with the minimum cost. That is, the transport costs $\sum_{k=1}^{K}\|\boldsymbol{h}_k^{(0)} - \boldsymbol{h}_k^{(1)}\|_2^2$ and $\|\boldsymbol{h}_G^{(0)} - \boldsymbol{h}_G^{(1)}\|_2^2$ are relatively small. Then, according to

$$\begin{aligned} &\sum_{k=1}^{K}\left\langle \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_G^{(0)}, \boldsymbol{h}_k^{(1)} - \boldsymbol{h}_G^{(1)} \right\rangle \\ &\quad = \frac{1}{2}\sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_G^{(0)} \right\|_2^2 + \frac{1}{2}\sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(1)} - \boldsymbol{h}_G^{(1)} \right\|_2^2 \\ &\qquad - \frac{1}{2}\sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_G^{(0)} - \boldsymbol{h}_k^{(1)} + \boldsymbol{h}_G^{(1)} \right\|_2^2 \\ &\quad \geq \frac{1}{2}\sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_G^{(0)} \right\|_2^2 + \frac{1}{2}\sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(1)} - \boldsymbol{h}_G^{(1)} \right\|_2^2 \\ &\qquad - \sum_{k=1}^{K}\left\| \boldsymbol{h}_k^{(0)} - \boldsymbol{h}_k^{(1)} \right\|_2^2 - K\left\| \boldsymbol{h}_G^{(0)} - \boldsymbol{h}_G^{(1)} \right\|_2^2 \end{aligned}$$

it is reasonable for the input data to satisfy the assumption above.

Theorem 1: Given Assumptions 1– 3, along the line (10), the PFC1 metric monotonically decreases to zero in $[0, 1]$.

Theorem 2: Given Assumptions 1 and 2, if the transport cost $\|\tilde{\boldsymbol{H}}(1) - \tilde{\boldsymbol{H}}(0)\|_F^2$ is sufficiently small, along the line (11), we have that the PFC2 metric monotonically decreases to zero in $[0, 1]$.

Intuitively, the features are NCC separable when they are concentrated and separated. That is to say, PFC1 and PFC2 can imply PFC3 implicitly. Thus, the PFC metrics decrease monotonically, indicating that the PFC happens. For PFC2, we need that the transport cost $\|\tilde{\boldsymbol{H}}(1) - \tilde{\boldsymbol{H}}(0)\|_F^2$ is sufficiently small to induce monotonic decrease. Gai and Zhang [28] observed that ResNet with weight decay attempts to learn an optimal transport map, and thus, a relatively small transport cost is reasonable.

However, for DNNs in practice, both the NC and geodesic curve assumption are not perfectly achieved. Thus, we prove the monotonicity of PFC in a more general setting.

Assumption 4 (ϵ-Geodesic Curve Assumption) At the terminal phase of training, the ResNet with weight decay learns the geodesic curve in the Wasserstein space approximately with an error ϵ, which means, according to the Benamou–Brenier formula

$$\int_0^1 \mathbb{E}\left[\left\| \frac{d\hat{\boldsymbol{h}}_{k,i}(t)}{dt} \right\|_2^2\right] dt \leq \int_0^1 \mathbb{E}\left[\left\| \frac{d\boldsymbol{h}_{k,i}^{(t)}}{dt} \right\|_2^2\right] dt + \epsilon \tag{12}$$

where $\hat{\boldsymbol{h}}_{k,i}^{(t)}$ is the feature in the forward propagation of ResNet and $\boldsymbol{h}_{k,i}^{(t)}$ is the feature satisfying Assumptions 1 and 2.

We need one more assumption on the connection between $\hat{\boldsymbol{h}}_{k,i}^{(t)}$ and $\boldsymbol{h}_{k,i}^{(t)}$.

Assumption 5: For $\forall k, i$, let $\boldsymbol{e}_{k,i}(t) = \hat{\boldsymbol{h}}_{k,i}^{(t)} - \boldsymbol{h}_{k,i}^{(t)}$ and $\boldsymbol{e}'_{k,i}(t) = \frac{d\boldsymbol{e}_{k,i}(t)}{dt}$. We assume that, for $\forall t \in [0, 1]$, $\boldsymbol{e}_{k,i}^{(t)}$ is independent of $\boldsymbol{h}_{k,i}^{(0)}$ and $\boldsymbol{h}_{k,i}^{(1)}$ across different i and k. Assume that $\boldsymbol{e}_{k,i}(0) = 0$, and $\mathbb{E}\boldsymbol{e}'_{k,i}(t) = \mathbb{E}\boldsymbol{e}_{k,i}(t) = 0\ \forall t \in [0, 1]$.

Remark 2: The source of $e_{k,i}^{(t)}$ can be complex. An intuitive illustration may be the inaccurate optimization method such as the SGD, which injects noise into features across layers. Thus, it is reasonable to assume that $e_{k,i}^{(t)}$ is independent of the input to the ResNet $h_{k,i}^{(0)}$ and the ideal feature of the last layer $h_{k,i}^{(1)}$.

Theorem 3: Given Assumptions 4 and 5, suppose that there exists a constant R such as $\|h_{k,i}^{(t)}\|_2^2 \le R^2\ \forall k, i, t$. Denote

$$V(t) \triangleq 2(1-t)\mathbb{E}\left[\left\|h_{k,i}^{(0)} - h_k^{(0)}\right\|_2^2\right] \cdot \left(\mathbb{E}\left[\left\|\tilde{h}_k^{(t)}\right\|_2^2\right] + (1-t)\mathbb{E}\left[\left\langle \tilde{h}_k^{(t)}, \tilde{h}_k^{(1)} - \tilde{h}_k^{(0)}\right\rangle\right]\right)$$

then for $\forall\epsilon$ satisfying $0 < \epsilon < (V(t))/((4 + 4/\sqrt{\delta})R^2)$, it holds that PFC1 metric monotonically decreases on a set of t with probability at least $1 - \delta$.

C. Multilayer UFM

Analyzing the NC phenomenon directly in DNNs is challenging due to the intricate interactions across multiple layers. Consequently, various studies have opted for simplification using UMF, focusing solely on the last-layer features and the classifier. The UFM regards the last-layer features as free variables and optimizes features and the classifier under loss function and regularizer or constraint. The global optimality achieved by UFM aligns with the NC phenomenon.

While UFM is a reasonable surrogate model for understanding NC, it has inherent limitations. First, UFM breaks the relation between the initial data and features, even though the initial data influences the collapsed features [44]. It neglects the structure of DNNs, especially the depth, which holds practical significance [56]. Next, for symmetry, UFM usually uses the regularizer or constraint on $\|H\|_F$ and $\|W\|_F$ simultaneously. Weight decay is commonly applied to all parameters without explicitly constraining the features. The formulation of UFM is quite idealized. In addition, we usually cannot reach perfect NC in practice. To overcome the limitations of UFM, we propose an MUFM based on the geodesic curve assumption.

In addition to the last-layer features, we treat all features $\{H^l\} = \{[h_{1,1}^l, \ldots, h_{1,n}^l, h_{2,1}^l, \ldots, h_{K,n}^l]\}$, $l = 1, \ldots, L$, as optimization variables in MUFM and connect them by the optimal transport regularizer, which is a lower bound of the weight decay as mentioned in Section II-C. Formally, recall that the general form of a ResNet block in forward propagation is

$$h^{(l+1)} = h^{(l)} + v_l\left(h^{(l)}\right), \quad l \in \{0, 1, \ldots, L-1\}$$

where $v_l(\cdot) = \sigma(W_l \cdot)$ is a shallow network, $\sigma(\cdot)$ is the ReLU function, and $h^{(l)}$ is the input of the lth ResNet block. Let μ_l denote the distribution of $h^{(l)}$ and assume that $\|h^{(l)}\|_2$ is upper bounded. Then, there exists a constant C such that

$$C\sum_{l=0}^{L-1}\left(\|W_l\|_F^2\right) \ge \sum_{l=0}^{L-1}\|v_l\|_{L^2(\mu_l)}^2 = \frac{1}{Kn}\sum_{l=0}^{L-1}\sum_{k=1}^{K}\sum_{i=1}^{n}\left\|h_{k,i}^{l+1} - h_{k,i}^l\right\|_2^2.$$

In particular, if the input distribution μ_l is symmetric and isotropic with zero mean (e.g., standard multivariate normal distribution), then the optimal transport regularizer is equivalent to the weight decay up to a constant factor. Actually

$$\begin{aligned}\|v_l\|_{L^2(\mu_l)}^2 &= \mathbb{E}_{h^{(l)}}\left[\left\|\sigma\left(W_l h^{(l)}\right)\right\|_2^2\right] \\ &= \frac{1}{2}\mathbb{E}_{h^{(l)}}\left[\left\|W_l h^{(l)}\right\|_2^2\right] \\ &= \frac{1}{2}\|W_l\|_F^2.\end{aligned}$$

Thus, it is reasonable to replace the weight decay on former layers with the optimal transport regularizer and only consider the weight decay on classifier W. Consider the following optimization problem:

$$\begin{aligned}\min_{W, H^1, \ldots, H^L}\ & \mathcal{L}_{\text{CE/MSE}}\left(W, H^L, Y\right) + \frac{\lambda_W}{2K}\|W\|_F^2 \\ & + \frac{\lambda}{2Kn}\sum_{l=0}^{L-1}\sum_{k=1}^{K}\sum_{i=1}^{n}\|h_{k,i}^{l+1} - h_{k,i}^l\|_2^2 \\ \text{s.t.}\ & h_{k,i}^0 = x_{k,i} \quad \forall k = 1, \ldots, K,\ i = 1, \ldots, n\end{aligned} \tag{13}$$

where

$$\mathcal{L}_{\text{CE}}\left(W, H^L, Y\right) = \frac{1}{Kn}\sum_{k=1}^{K}\sum_{i=1}^{n}\text{CE}\left(W h_{k,i}^L, y_k\right)$$

$$\mathcal{L}_{\text{MSE}}\left(W, H^L, Y\right) = \frac{1}{2Kn}\left\|W H^L - Y\right\|_F^2$$

$\otimes$ denotes the Kronecker product, $\lambda_W, \lambda > 0$ are the hyperparameters of regularizers, and $\mathbf{Y} = \mathbf{I}_K \otimes \mathbf{1}_n^\top \in \mathbb{R}^{K \times Kn}$ for the MSE loss.

When minimizing the optimal transport regularizer, if the last-layer features are determined, the features of intermediate layers should lie in a straight line between the data and the last-layer features because all features are free variables. Thus, minimizing the optimal transport regularizer can be reduced to minimize the distance between the data and the last-layer features, as shown in the following theorem.

Theorem 4: The problem in (13) is equivalent to

$$\min_{W, H}\mathcal{L}_{\text{CE/MSE}}(W, H, Y) + \frac{\lambda_W}{2K}\|W\|_F^2 + \frac{\lambda}{2Kn}\|H - X\|_F^2 \tag{14}$$

where X is the input data matrix and H is the last-layer feature matrix for simplicity.

Proof: $\forall k, i$, when $h_{k,i}^0$ and $h_{k,i}^L$ are fixed, we have

$$\begin{aligned}\sum_{l=0}^{L-1}\left\|h_{k,i}^{l+1} - h_{k,i}^l\right\|_2^2 &\ge \left\|\sum_{l=0}^{L-1}\left(h_{k,i}^{l+1} - h_{k,i}^l\right)\right\|_2^2 \\ &= \left\|h_{k,i}^L - h_{k,i}^0\right\|_2^2\end{aligned} \tag{15}$$

and the equality is achieved only when $\forall l$

$$h_{k,i}^{l+1} - h_{k,i}^l = \frac{1}{L}\left(h_{k,i}^L - h_{k,i}^0\right) \tag{16}$$

which concludes the proof. ■

We call this model (14) the MUFM. It is easy to show that, while taking data distribution into account, the global optima of MUFM are not consistent with NC. In fact, (14) can be rewritten as

$$\min_{\boldsymbol{W},\boldsymbol{H}} \mathcal{L}_{\mathrm{CE/MSE}}(\boldsymbol{W},\boldsymbol{H},\boldsymbol{Y}) + \frac{\lambda_W}{2K}\|\boldsymbol{W}\|_F^2 + \frac{\lambda}{2Kn}\|\boldsymbol{H}\|_F^2 - \frac{\lambda}{Kn}\mathrm{Tr}\left(\boldsymbol{X}\boldsymbol{H}^\top\right). \tag{17}$$

By the results of UFM, the solution of (17) without the last term $\lambda/(Kn)\mathrm{Tr}(\boldsymbol{X}\boldsymbol{H}^\top)$ is the NC solution. However, $\lambda/(Kn)\mathrm{Tr}(\boldsymbol{X}\boldsymbol{H}^\top)$ makes $\boldsymbol{H}$ align with $\boldsymbol{X}$. To exhibit the collapse phenomenon in MUFM theoretically, we have the following.

Theorem 5: Suppose that the initial data $\boldsymbol{X}$ are noncollapsed, and then, the optimal solutions of (14) are also noncollapsed. In addition, if the loss function is chosen to be MSE, the minimizer with respect to $\boldsymbol{W}$ has a closed-form expression $\boldsymbol{W}^*(\boldsymbol{H}) = \boldsymbol{Y}\boldsymbol{H}^\top(\boldsymbol{H}\boldsymbol{H}^\top + n\lambda_W\boldsymbol{I}_d)^{-1}$, which is a function of $\boldsymbol{H}$. If we substitute $\boldsymbol{W}^*(\boldsymbol{H})$ for $\boldsymbol{W}$, then there exists a large enough constant $C > 0$, and for any $\lambda > C$, the minimizer $\boldsymbol{H}_{1/\lambda}$ of (14) satisfies $\mathcal{PFC}_1(\boldsymbol{H}_{1/\lambda}) < \mathcal{PFC}_1(\boldsymbol{X})$, which means that the features are more collapsed than the data.

The proof of Theorem 5 is analogous to the theorems in [24]. For completeness, we provide it in Appendix D (Supplementary Material). The coefficient of the optimal transport regularizer plays an essential role in the tradeoff between the data and ETF. If the coefficient is small, the solutions will be close to those of UFM, i.e., the simplex ETF. If it is large, the solutions will be close to the data but still more collapsed than the data. We give empirical results to show the collapsed degree of solutions and the tradeoff between the simplex ETF and data in Section IV-B. When the coefficient is large or small, we provide two corollaries in Appendix D (Supplementary Material) to connect MUFM and PFC.

IV. Experiments

Here, we present the empirical evidence supporting the PFC conjecture when training ResNet on various datasets. We also show that these PFC metrics monotonically decrease along a straight line under the geodesic curve assumption. Then, we empirically show the distinctions between the optimal solutions of UFM and MUFM. The optimal solution of MUFM is not the simplex ETF, and there exists a tradeoff between data and simplex ETF by controlling the coefficient λ of the optimal transport regularizer. Finally, we extend PFC to ViT and provide some empirical evidence.

A. PFC Conjecture

To compare PFC metrics among different layer features, we need to keep the features in the same dimension (dim.). In this section, we first train a fully connected (FC) ResNet whose intermediate dimension is set to 1000 on the MNIST dataset. For the other datasets, such as Fashion MNIST, CIFAR10, STL10, and CIFAR100, we train the general convolutional (Conv) ResNets. These general ResNets gradually reduce the dimension to extract features, and we focus our investigation solely on the final four or five blocks whose input features are of the same dimension. The details of network architectures are presented in Table I. The architectures are selected to satisfy that the empirical effective depth equals the actual depth since we only characterize PFC before effective depth. Thus, the last-layer features are NCC separable, indicating NC emerges. Additionally, to further validate the consistency of our findings, we train additional ResNets with varying depths on each dataset, as detailed in Appendix F (Supplementary Material).

TABLE I
Architectures of ResNet for Various Datasets

Dataset	Data dim.	K	Intermediate dim.	# of blocks	Layer
MNIST	$1 \times 28 \times 28$	1	1000	10	FC
Fashion MNIST	$1 \times 28 \times 28$	2	$32 \times 14 \times 14$	5	Conv
			$64 \times 7 \times 7$	4	
CIFAR10	$3 \times 32 \times 32$	3	$64 \times 16 \times 16$	2	Conv
			$128 \times 8 \times 8$	3	
			$256 \times 4 \times 4$	4	
STL10	$3 \times 96 \times 96$	4	$64 \times 48 \times 48$	5	Conv
			$128 \times 24 \times 24$	5	
			$256 \times 12 \times 12$	5	
			$512 \times 6 \times 6$	5	
CIFAR100	$3 \times 32 \times 32$	4	$64 \times 32 \times 32$	3	Conv
			$128 \times 16 \times 16$	3	
			$256 \times 8 \times 8$	4	
			$512 \times 4 \times 4$	4	

Similar to the optimization setting in [15], we minimize the CE loss of each ResNet using SGD with 0.9 momentum. The weight decay is set to 5×10^{-4}, and the batch size is 128. We train the network for 350 epochs. The initial learning rate is set to 0.005 for MNIST and 0.01 for other datasets, and it is divided by 10 at 117th and 233rd epochs.

To demonstrate the PFC conjecture empirically, we consider the following metrics in each layer to measure PFC.

(PFC1) Variability Collapse:

$$\mathcal{PFC}_1\left(\boldsymbol{H}^l\right) := \frac{\mathbb{E}\left[\left\|\boldsymbol{h}_{k,i}^l - \boldsymbol{h}_k^l\right\|_2^2\right]}{\mathbb{E}\left[\left\|\boldsymbol{h}_k^l - \boldsymbol{h}_G^l\right\|_2^2\right]}. \tag{18}$$

(PFC2) Convergence to Simplex ETF:

$$\mathcal{PFC}_2\left(\boldsymbol{H}^l\right) := \left\|\frac{\tilde{\boldsymbol{H}}^{l\top}\tilde{\boldsymbol{H}}^l}{\left\|\tilde{\boldsymbol{H}}^{l\top}\tilde{\boldsymbol{H}}^l\right\|_F} - \boldsymbol{E}\right\|_F \tag{19}$$

where $\tilde{\boldsymbol{H}}^l := [\boldsymbol{h}_1^l - \boldsymbol{h}_G^l, \ldots, \boldsymbol{h}_K^l - \boldsymbol{h}_G^l] \in \mathbb{R}^{d\times K}$ is the centered class mean matrix and $\boldsymbol{E} = 1/((K-1)^{1/2})(\boldsymbol{I}_K - 1/K\boldsymbol{1}_K\boldsymbol{1}_K^\top)$.

(PFC3) NCC Accuracy:

$$\mathcal{PFC}_3\left(\boldsymbol{H}^l\right) := \frac{1}{nK}\sum_{k=1}^{K}\sum_{i=1}^{n}\mathbb{I}\left[\arg\min_{c\in[K]}\left\|\boldsymbol{h}_{k,i}^l - \boldsymbol{h}_c^l\right\|_2 = k\right]. \tag{20}$$

We illustrate the evolution of the PFC metrics during training on various datasets (Fig. 2). To show the degree of collapse at each layer more clearly, we plot the PFC metrics of each layer at the final epoch (Fig. 3). The horizontal axis represents their relative position, scaled to the range of

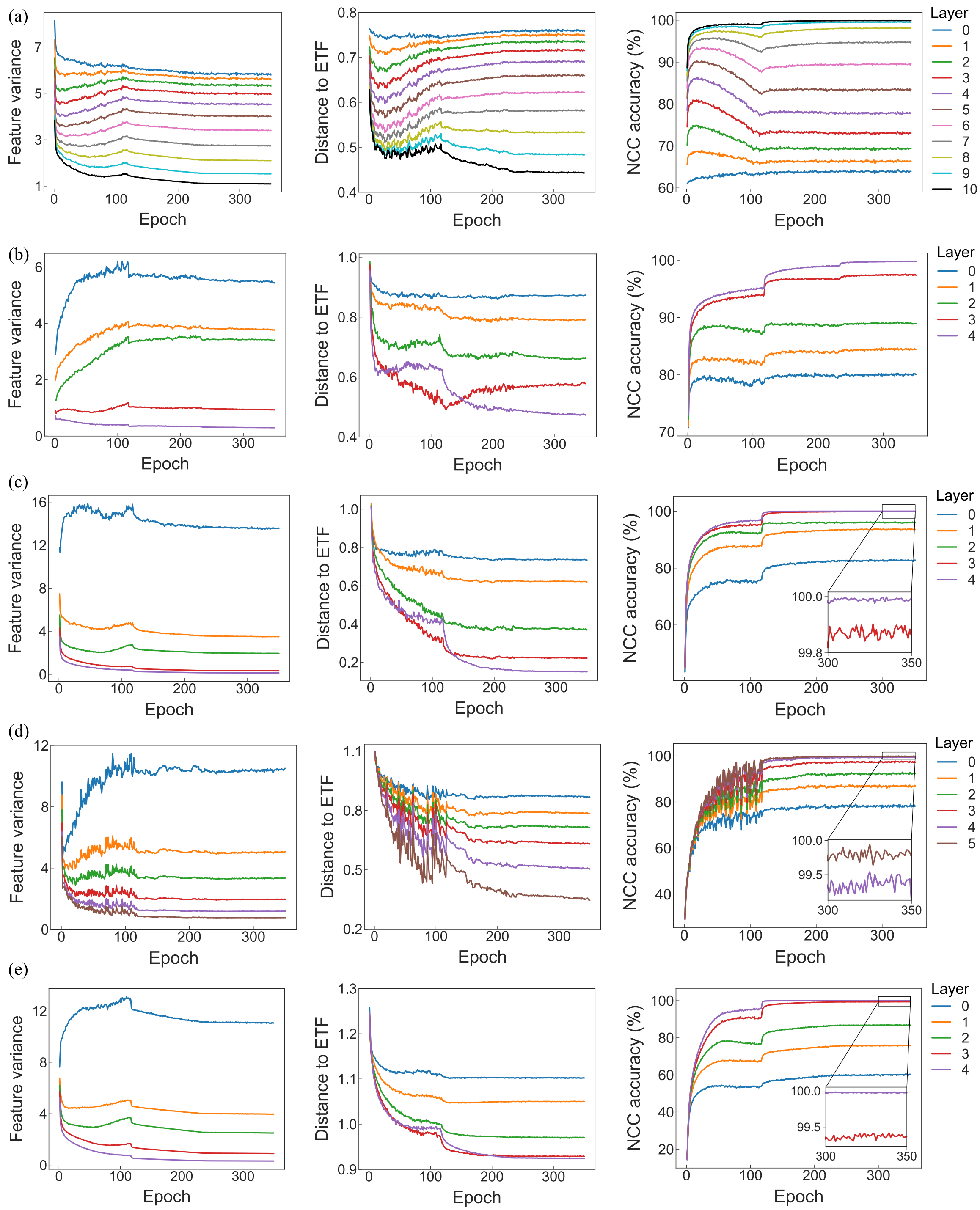

Fig. 2. Features of ResNet exhibiting PFC on various datasets. In each row, the PFC metrics of different layers on (a)–(e) MNIST, fashion MNIST, CIFAR10, STL10, and CIFAR100 are plotted, respectively; "layer0" represents the input features.

$[0, 1]$. Formally, for l-layer features $\{\boldsymbol{h}_{k,i}^l\}$, their coordinate is calculated as follows:

$$\frac{\sum_{l=0}^{l-1}\sum_{k=1}^{K}\sum_{i=1}^{n}\left\|\boldsymbol{h}_{k,i}^{l+1}-\boldsymbol{h}_{k,i}^{l}\right\|_2}{\sum_{l=0}^{L-1}\sum_{k=1}^{K}\sum_{i=1}^{n}\left\|\boldsymbol{h}_{k,i}^{l+1}-\boldsymbol{h}_{k,i}^{l}\right\|_2}.$$

Additionally, the geodesic curve assumption allows us to predict the metrics along the line $\boldsymbol{h}_{k,i}^{(t)} = (1-t)\boldsymbol{h}_{k,i}^{(0)} + t\boldsymbol{h}_{k,i}^{(1)}, t \in [0, 1]$, which connects the input features $\{\boldsymbol{h}_{k,i}^{(0)}\}$ and the last-layer features $\{\boldsymbol{h}_{k,i}^{(1)}\}$. For each $\{\boldsymbol{h}_{k,i}^{(t)}\}, t \in [0, 1]$, we compute the PFC metrics and connect them (blue curve in Fig. 3). Empirical evidence supports the PFC conjecture when training ResNets. More concretely, before the effective depth, features of each layer tend to collapse to their class means progressively, (centered) class means of each layer tend to collapse to the simplex ETF progressively, and NCC accuracy rates of each layer tend to collapse to 1 progressively. Under the geodesic curve assumption, the metrics monotonically decrease along

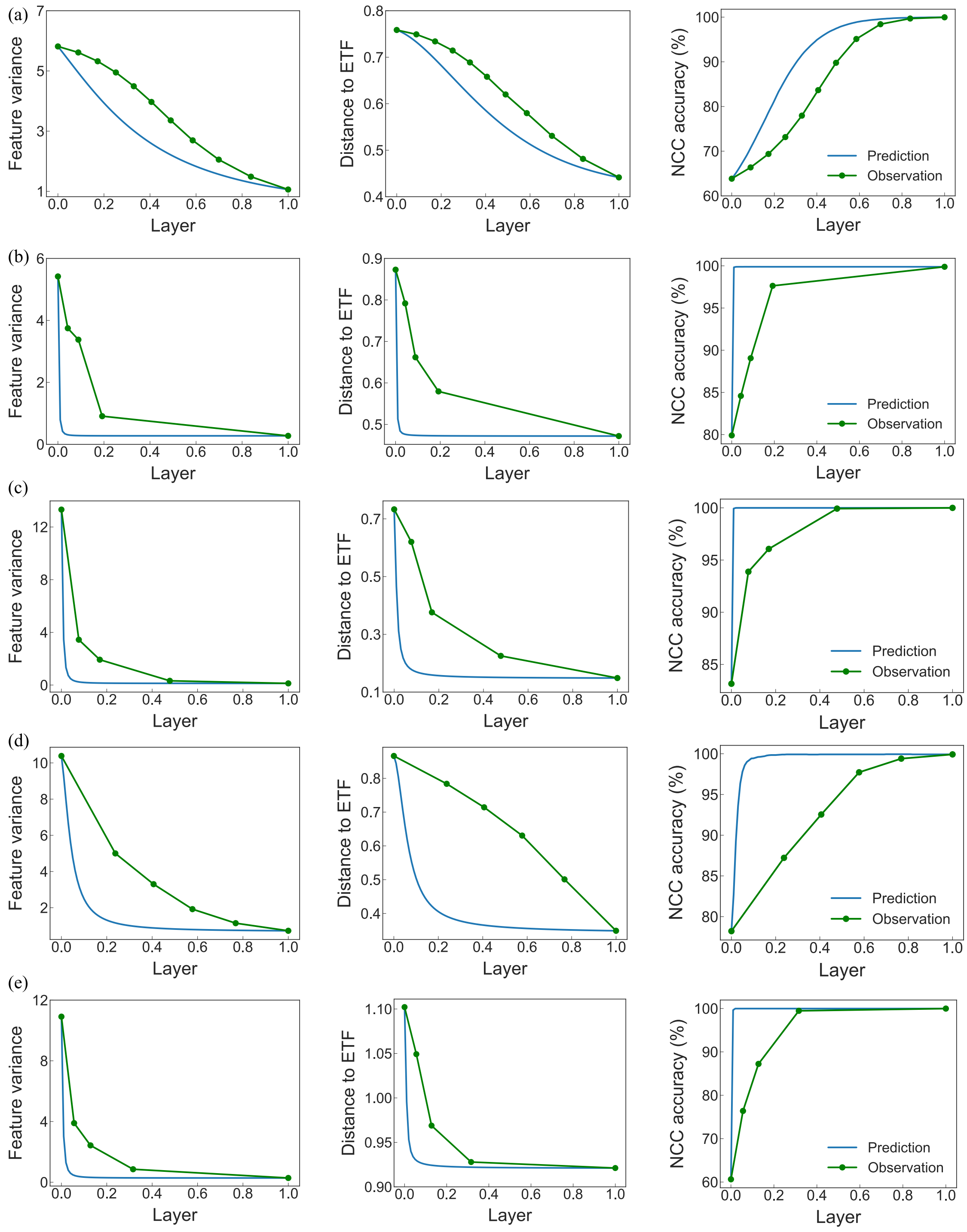

Fig. 3. PFC on various datasets at the final epoch. The green points are calculated by the features of different layers and connected by a green line. The blue curve represents the predicted values under the geodesic curve assumption. The abscissa represents the relative position where 0 corresponds to the input features. Both the observations and predictions are monotonic on various datasets (a)–(e) corresponding to the same datasets as in Fig. 2.

the straight line. For each layer, these metrics start at random initialization and converge to PFC at the final epoch.

B. Numerical Experiments in MUFM

In this section, we numerically solve both MUFM and UFM. The solutions of both UFM and MUFM exhibit the NC phenomenon, while the NC in MUFM is slighter (Fig. 4). Furthermore, MUFM shows the tradeoff between the simplex ETF and data while changing the coefficient of optimal transport regularizer λ (Fig. 5).

Consider MUFM (14) in the setting $k = 5, d = 20, n = 100, \lambda_{\boldsymbol{W}} = 0.005, \lambda = 0.001$, and $\boldsymbol{W}$ and $\boldsymbol{H}$ are initialized with the standard normal distribution. MSE is used as the loss function. The data $\{\boldsymbol{x}_{k,i}\}_{i=1}^{n}$ for each class k are sampled from a Gaussian random vector $\mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{I}_d)$, where $\boldsymbol{\mu}_k \in \mathbb{R}^d$ is uniformly

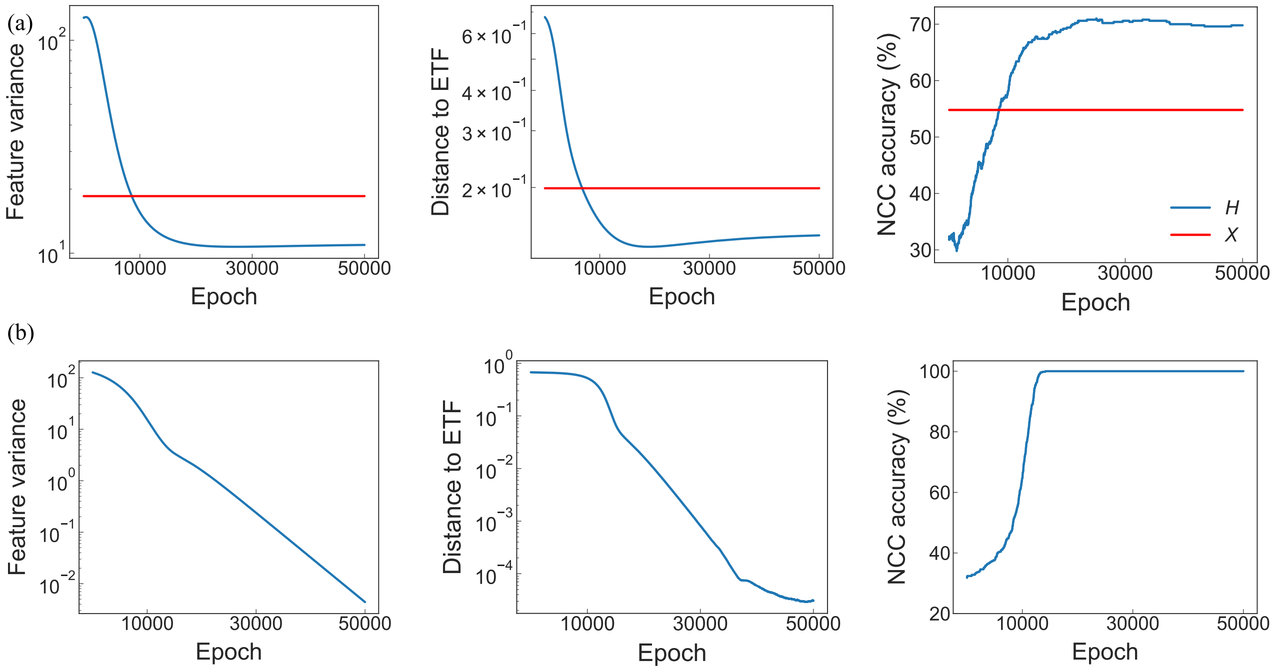

Fig. 4. Difference between MUFM and UFM in terms of the global optimality. (a) Collapse degree of the solution of MUFM, measured by the PFC metrics. The red line is the collapse degree of data. (b) Collapse degree of the solution of UFM, measured by the PFC metrics (log scale). The solution of UFM exhibits NC, but MUFM does not.

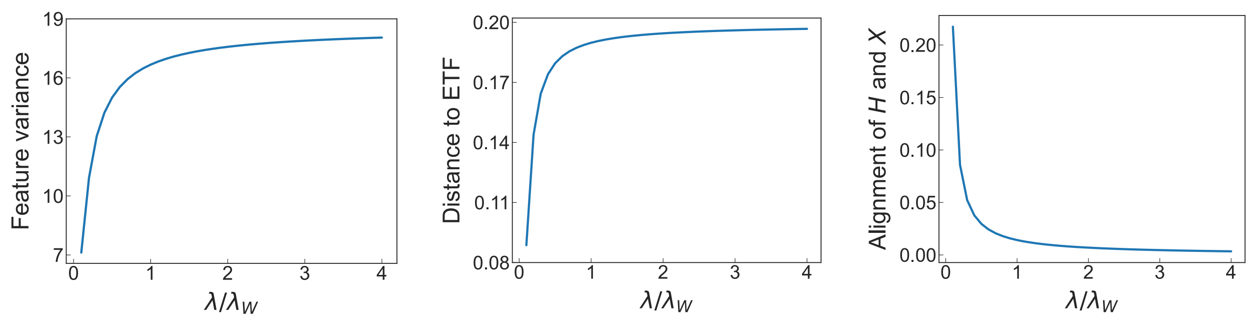

Fig. 5. Tradeoff in MUFM. The abscissa is the ratio $\frac{\lambda}{\lambda_{\boldsymbol{W}}}$, and we keep the value of $\lambda_{\boldsymbol{W}}$ and only vary λ. Increasing λ makes the solution of MUFM away from the NC solution (left middle). Increasing λ makes the solution of MUFM close to the data (right).

sampled from the unit sphere. UFM is in the same setting without sampling $\boldsymbol{X}$. We solve the problems by plain gradient descent with a learning rate 0.1 for 5×10^4 epochs. We then measure the solution $\boldsymbol{H}$ specifically by PFC metrics to show if it is consistent with NC (original NC includes self-duality, i.e., the alignment of the classifier and the last-layer features; we will not measure this additionally). We can see that the solution of UFM is much closer to the NC solution (each metric is in a smaller magnitude), but the solution of MUFM is not (Fig. 4). Theorem 5 tells us that $\mathcal{PFC}_1(\boldsymbol{H}) < \mathcal{PFC}_1(\boldsymbol{X})$ for large enough λ, and we compute all PFC metrics of the data $\boldsymbol{X}$ and show that the collapse degree (not only $\mathcal{PFC}_1$, but all PFC metrics) of solution $\boldsymbol{H}$ is larger than data empirically.

To show the tradeoff between data and simplex ETF, we gradually vary the optimal transport regularizer coefficient λ from 0.0005 to 0.02 while keeping other parameters constant. For each value of λ, we solve the MUFM as above and measure the collapse degree of $\boldsymbol{H}$ by PFC metrics. In addition, we measure the alignment of the solution $\boldsymbol{H}$ and the data $\boldsymbol{X}$ by

$$\left\| \frac{\boldsymbol{H}}{\|\boldsymbol{H}\|_F} - \frac{\boldsymbol{X}}{\|\boldsymbol{X}\|_F} \right\|_F .$$

We illustrate that for small values of the coefficient λ, the solution tends to be close to the simplex ETF (Fig. 5). Conversely, for large values of λ, the solution moves away from the ETF and becomes closer to the initial data.

C. PFC in ViT

We further investigate PFC with ViT [33]. Due to the residual connections in ViT, the forward propagation of ViT can also be viewed as a discretization of the continuity equation. Thus, the ViT with weight decay approximates the Wasserstein geodesic curve induced by the optimal transport map at the terminal phase of training. This suggests that ViT, similar to ResNet, exhibits PFC. To empirically validate this, we train a

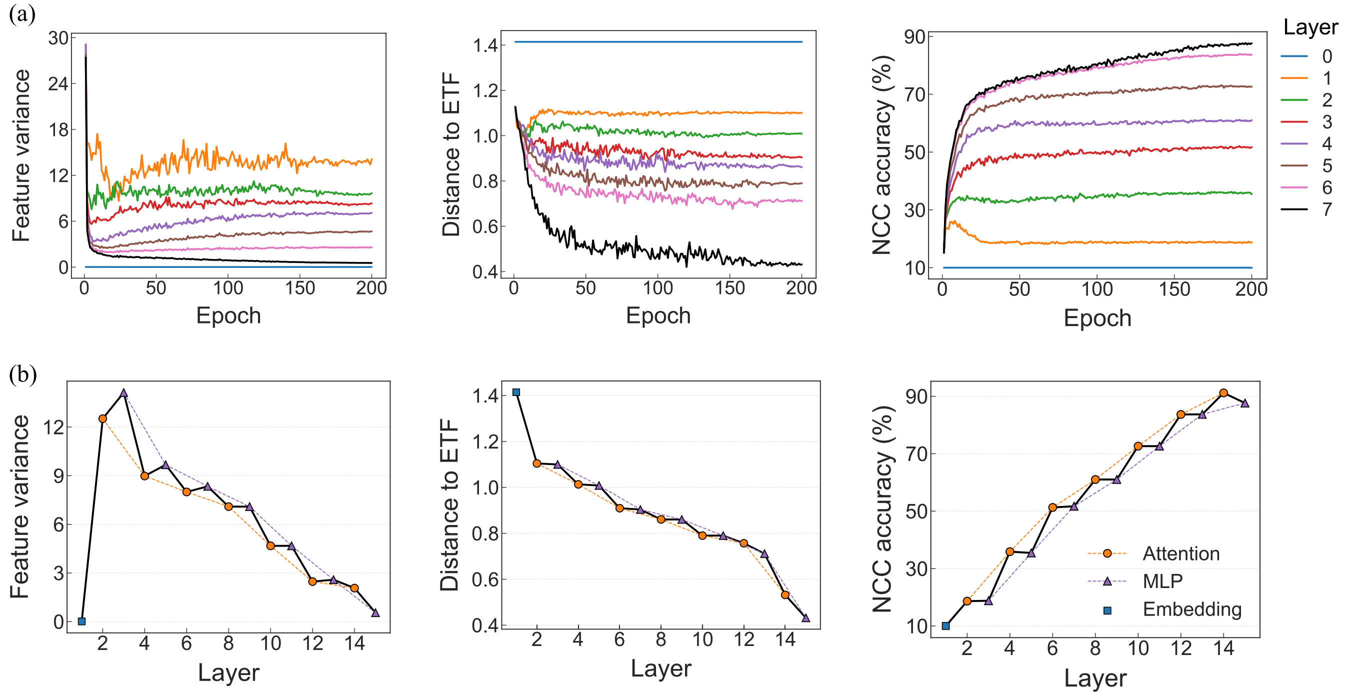

Fig. 6. Features of ViT exhibiting PFC on CIFAR10. (a) Evolution of the PFC metrics for the class token across ViT layers during training; "layer0" represents the embedding layer (input feature). (b) PFC metrics computed at the final epoch for the attention and MLP components within each ViT block. The PFC metrics decline more sharply in the attention layers than in the MLP layers.

ViT from scratch with 7 blocks and 12 heads on CIFAR10. The input images are divided into 64 patches, each mapped to a 384-D embedding, with an additional class token appended for classification. The model is optimized using the AdamW optimizer with a cosine annealing learning rate scheduler. We use the class token of each layer to calculate the PFC metrics.

Unlike ResNet, ViT performs classification through the class token, and all samples in ViT share the same initialization for the class token in the embedding layer. Consequently, the PFC metrics of the embedding layer remain unchanged during training, especially $\mathcal{PFC}_1 = 0$. We illustrate the evolution of PFC metrics during training in Fig. 6(a), showing that, except for the embedding layer, all PFC metrics decrease monotonically with increasing depth, consistent with the PFC conjecture. Notably, due to the identical initialization of the class token in the embedding layer, $\mathcal{PFC}_1$ is fixed at zero there, and it is only in the subsequent layers that $\mathcal{PFC}_1$ begins to decline gradually.

To further analyze the behavior, we plot the PFC metrics of each layer at the final training epoch in Fig. 6(b). Each block in ViT consists of a multihead attention layer followed by an MLP layer. We separately compute PFC metrics for these components and find that within the class token representations, the attention layer contributes more significantly to classification (indicated by the PFC metrics) than the MLP layer. This suggests that the attention mechanism is dominant in shaping the class token features toward NC, whereas the MLP layers primarily refine and stabilize these representations. In addition, we present more results of ViTs and their variants on CIFAR100 and ImageNet in Appendix E (Supplementary Material).

V. Related Work

A. NC and (Extended) UFM

The NC phenomenon has been described in [15] and [16] and proved to be emerging by UFM under various settings. Han et al. [16] and Mixon et al. [34] proved that the dynamics of SGD converge to the NC solutions. Several studies proved that the global minimizers are NC solutions [35], [36], [37], [38], [39], [40], [41], [42], [43]. Moreover, some studies [40], [41], [42], [43] proved that UFM has benign loss landscape. Beyond UFM, Tirer and Bruna [39] extended UFM to two layers and characterized its global optimality. Dang et al. [53] extended UFM to multiple linear layers and characterized the global optimality. Súkeník et al. [46] extended UFM to arbitrary nonlinear layers and proved that after a certain layer, the optimal solutions for binary classification exactly exhibit NC. Súkeník et al. [57] further extended binary classification to multiple classification, proving that NC2 in the last layer is not optimal in this setting. Since UFM does not consider the effect of data, in our article, we propose a more general model MUFM, which models the cases when NC is inaccurate. Tirer et al. [24] also proposed a new model to analyze the "near collapse" situation, which is similar to MUFM. MUFM is constructed more reasonably by replacing weight decay with the optimal transport regularizer. We claim that the regularizer on $\boldsymbol{H}$ is unnecessary to model DNNs, which is not used in practice. Yang et al. [44] also considered the role of the initial data in NC, showing that the collapsed features corresponding to each label can exhibit fine-grained structures determined by the initial data. This provides experimental evidence to

MUFM for modeling the alignment between the input data and features.

B. NC at the Intermediate Layers

The NC phenomenon only characterizes the last-layer features. Several studies tried to extend it to intermediate layers. Many empirical results show that NC1 emerges at intermediate layers but progresses more weakly than the last layer [20], [21], [22], [23], [24], [25]. Ben-Shaul and Dekel [22] and Galanti et al. [23] also provided a similar result of NC4. These empirical works did not explore all NC properties. Rangamani et al. [45] extended all NC properties to intermediate layers, including NC3 (feature-weight alignment). However, Rangamani et al. [45] did not characterize the evolution of the NC properties; instead, it showed that there exists a hidden layer beyond which all subsequent layers exhibit the NC properties. They provided the NC properties after the effective depth. Parker et al. [58] provided similar empirical results in different settings to characterize the intermediate layers. Our PFC conjecture focuses on the layers of ResNet before effective depth. Moreover, we show the monotonicity of collapse degree and suggest some theoretical understandings.

Tirer et al. [24] also provided a theoretical result that the PFC1 metric monotonically decreases along a gradient flow. Thus, the PFC1 metric will decrease across depth when the network is optimized layerwise (a new layer is added on top of the trained network each time, i.e., cascade learning). However, this theoretical result does not match the more common empirical observation perfectly [20], [21], [22], [23], whose training framework is end-to-end. Our result shows this monotonic decrease of PFC1 metric across depth directly for end-to-end training with some mild assumptions.

VI. Conclusion and Discussion

NC is an intriguing phenomenon observed during the training of DNNs, revealing insights into the geometry of the last-layer features and classifier, thereby elucidating the efficacy of DNNs for classification tasks. In this work, we extend the NC phenomenon to the intermediate layers of ResNet and propose a novel conjecture termed PFC indicated by PFC1–PFC3. We provide empirical evidence to support the PFC conjecture. In theory, we introduce the geodesic curve assumption and a relaxed version closely aligned to the deep networks in practice. We prove that the metrics of PFC indeed decrease monotonically across depth at the terminal phase of training under some assumptions. The PFC conjecture suggests that the ResNet can progressively concentrate the features and separate the class means to achieve classification problems.

The geodesic curve assumption also sheds light on extending UFM to MUFM. Considering the intermediate layer, we propose a new surrogate model, MUFM, to understand NC and PFC. This model establishes a connection between the data and the last-layer features through the optimal transport regularizer. Then, the optimal solution of MUFM is inconsistent with NC, but the features are more concentrated than the input data. The optimal transport regularizer coefficient λ can control the distance between the solution and the data. Compared to UFM, MUFM models the terminal phase of DNN more comprehensively.

Further research could explore more theoretical aspects of MUFM. Current PFC conjecture and theoretical understanding focus on ResNet. It will be interesting to characterize the forward propagation of other DNN architectures. Additionally, we hope that the PFC conjecture could inspire the design of new loss functions or architectures aimed at improving classification performance.

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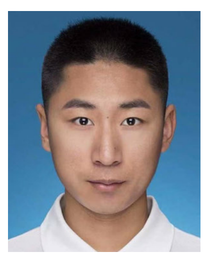

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